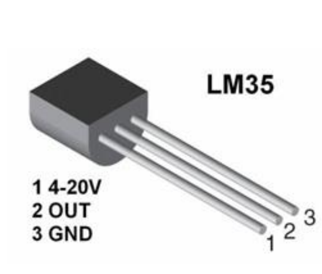
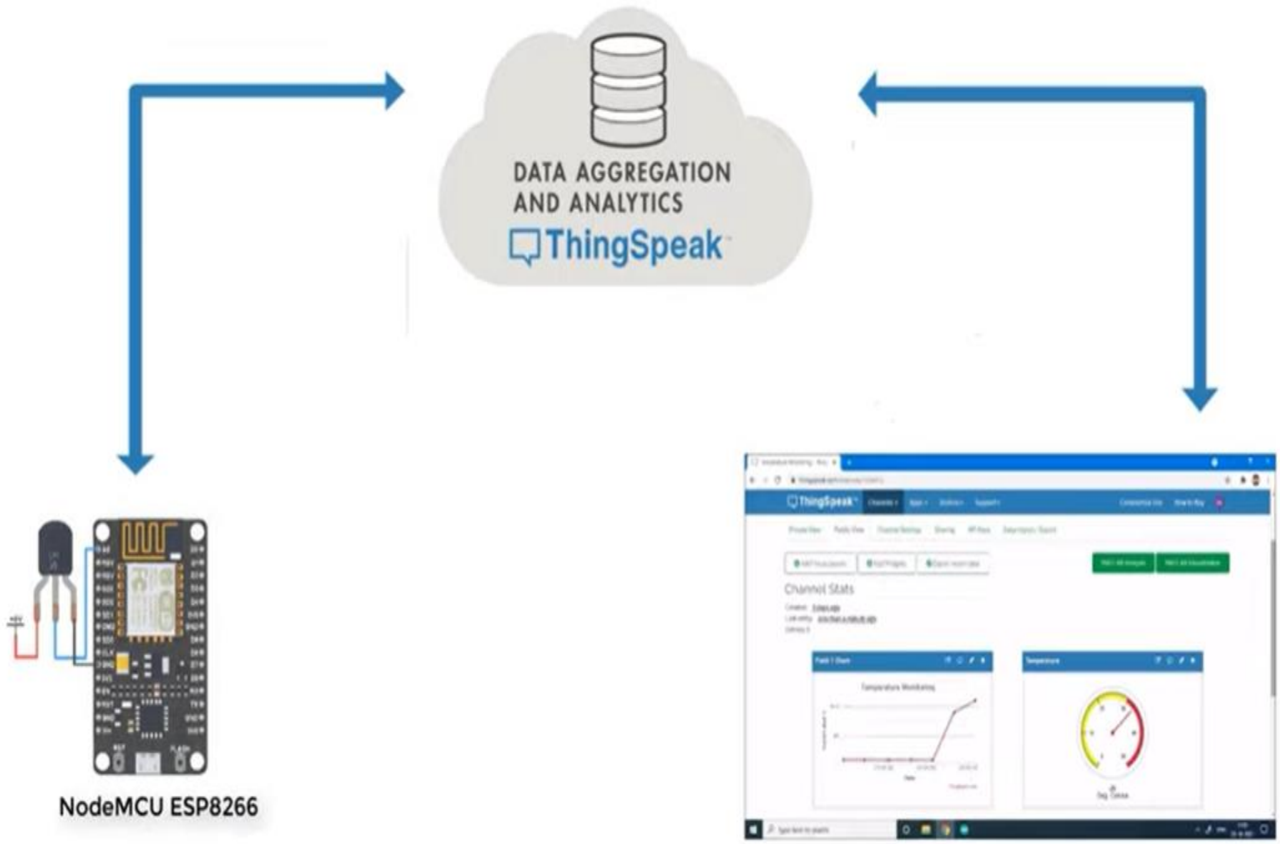
Lab 7

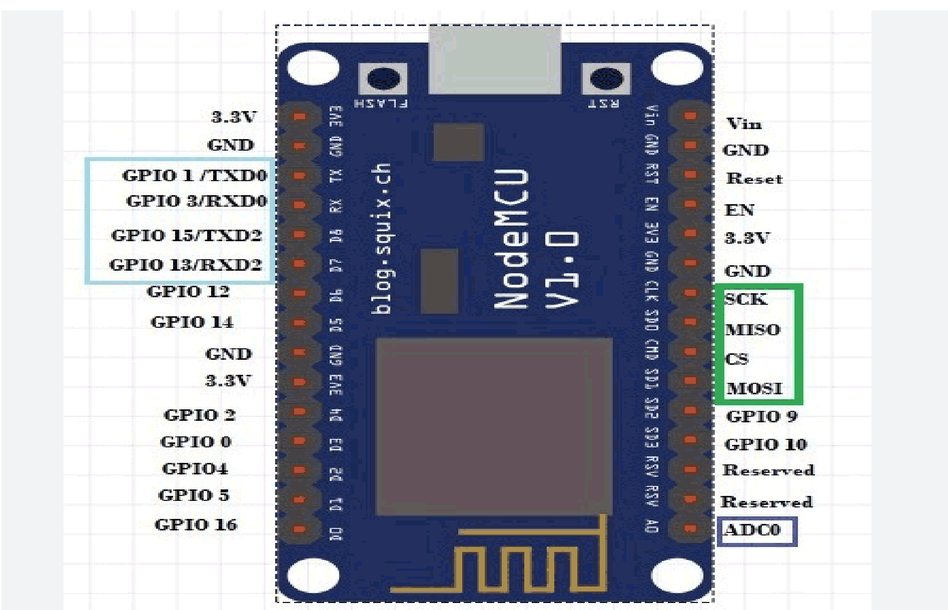
Send Temperature Sensor Data from NodeMCU ESP8266 to ThingSpeak Cloud Platform

The project will break down into the following steps:

1. Collect components (NodeMCU ESP8266, LM35 temperature sensor)
2. We'll connect NodeMCU ESP8266 with laptop using MicroUSB cable.
3. And then install ESP8266 specific software packages using board manager in arduino software.
4. This will let us select NodeMCU1.0 ESP-12E module while programming NodeMCU in tools section.
5. Connect components according to the following figure (1 number pin of lm35 connect to NodeMCU 3V pin, 2 number pin connect to A0 pin, and 3 number pin connect to ground of NodeMCU).
6. Setup ThingSpeak Account
7. Create ThingSpeak Channel
8. Download and Install ThingSpeak libraries
9. Write Arduino program for NodeMCU ESP8266
10. Create ThingSpeak IoT Web Dashboard
11. Share Temperature Sensor Data anywhere in the world
12. If port is not available then install device driver software from Techshopbd (NodeMCU ESP8266 V3) website.




DATA AGGREGATION
AND ANALYTICS
ThingSpeak
Channel Stats
NodeMCU ESP8266


3.3V
GND
GPIO 1 /TXD0
GPIO 3/RXD0
GPIO 15/TXD2
GPIO 13/RXD2
GPIO 12
GPIO 14
GND
3.3V
GPIO 2
GPIO 0
GPIO4
GPIO 5
GPIO 16
FLASH
RST
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NodeMCU
V1.0
Vin
GND
Reset
EN
3.3V
GND
SCK
MISO
CS
MOSI
GPIO 9
GPIO 10
Reserved
Reserved
ADC0

```
#include <ESP8266WiFi.h>;
#include <WiFiClient.h>;
#include <ThingSpeak.h>;
const char* ssid = "";   // Your Network SSID
const char* password = "";       // Your Network Password
int val;
int pin = A0; // LM35 Pin Connected at A0 Pin

WiFiClient client;
unsigned long myChannelNumber = ; //Your Channel Number (Without Brackets)
const char * myWriteAPIKey = ""; //Your Write API Key
void setup()
{
  Serial.begin(9600);
  delay(10);
  WiFi.begin(ssid, password);
  ThingSpeak.begin(client);
}
void loop()
{
  val = analogRead(pin)*0.322265; // Read Analog values and Store in val variable
  Serial.print("Temperature: ");
  Serial.print(val);                 // Print on Serial Monitor
  Serial.println("*C");
  delay(1000);
  ThingSpeak.writeField(myChannelNumber, 1,val, myWriteAPIKey); //Update in ThingSpeak
  delay(100);
}
```